

Mohamed Abdelrahman

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[https://github.com/mohameds-dev/](https://github.com/mohameds-dev) mohamed-a-portfolio.vercel.app

EDUCATION

University of Houston

Aug 2023 – Dec 2026

Bachelor of Science in Computer Science

Houston, TX

- Major GPA: 3.50 / 4.00
- Relevant Coursework: Software Design, Algorithms and Data Structures, Database Systems, Operating Systems

EXPERIENCE

Director of Competitions

Jan 2025 – May 2025

Code[Coogs] - University Of Houston

Part Time - Houston, TX

- Organized and hosted 9 competitive programming events teaching data structures, algorithms, and complexity analysis to 30+ students
- Curated Kattis contests aligned with interview-style problem solving; facilitated live coding and code walkthroughs
- Collaborated with student leaders and faculty to iterate on formats and improve engagement

Coding Expert - AI Trainer

Aug 2023 – Jan 2025

Outlier.ai

Contract - Remote

- Developed solutions for fine-tuning code generation models in Python and C++, improving quality by 40% based on evaluation rubrics
- Participated in code reviews by assessing submissions and providing technical feedback to improve task quality by 30%

Software Engineering Intern

Aug 2022 – Jan 2023

bld.ai

Part Time - Remote

- Collaborated with 5 interns to develop a full-stack web application using React.js frontend and Django backend with PostgreSQL database
- Built and optimized REST APIs, database models, and data validation systems, reducing codebase bugs by 50%
- Shipped an MVP within one month using two-week sprint cycles; followed Agile practices (planning, stand-ups, retrospectives) and participated in code reviews

PROJECTS

TallyCart (In Progress) | *Django, Python, Flutter, EasyOCR, LLM Prompt Engineering*

- Developing full-stack web application with Django REST backend and Flutter application for shopping recommendations
- Building scalable database models and REST APIs to handle large volumes of receipt-derived data efficiently
- Integrating EasyOCR with LLM-powered postprocessing to convert raw receipt text into structured records via API endpoints

C++ Ordered Map Library | *C++17, CMake, LCOV, Valgrind, Catch2, GitHub Actions, TDD*

- Implemented ordered map data structure using linked list and hash table for O(1) lookups while preserving insertion order
- Achieved 97.3% code coverage across 2,246 assertions with Catch2; validated memory safety with Valgrind
- Designed STL-compatible interface with iterator support and exception safety using modern C++17
- Set up CI/CD with GitHub Actions and CMake for automated builds and tests

ACHIEVEMENTS

Egyptian Collegiate Programming Contest'22 Finalist — ACM-ICPC

- Placed 73rd in the 2022 ICPC Egypt Regional (ECPC) after advancing through university and national qualification rounds from over 1800 nationwide teams, solving advanced algorithmic problems in C++ with a 3-member team.

PROGRAMMING SKILLS

Backend: C++, Python, Django, SQL, REST APIs

Frontend: JavaScript, HTML/CSS, React, Flutter

DevOps & Tools: Git, Docker, GitHub Actions, CI/CD, Linux, Bash, Coverage